

Brice Peres

PhD candidate – Computer Graphics & Physical Simulation



Experience

Inria Elan team | PHD CANDIDATE – NUMERICAL STUDY OF NATURALLY-CURVED FIBRE ASSEMBLIES

Apr. 2026 – present | Grenoble, France

- › Numerical modelling and simulation of bundles of naturally-curved fibres (continuation of the work below).

Inria Elan team | PRE-PHD RESEARCHER – TOMOGRAPHIC HAIR-DATA FILTERING

Sept. 2025 – Mar. 2026 | Grenoble, France

- › Designed filtering methods for tomographic acquisitions of human hair, in preparation of the PhD project.

Dassault Systèmes | END-OF-STUDIES INTERNSHIP

Sept. 2024 – Mar. 2025 | Paris, France

- › Worked on the performance of one of Dassault Systèmes' main 3D engines, alongside the rendering team.

Corys | R&D INTERN – UNREAL ENGINE 5

May 2023 – Sept. 2023 | Grenoble, France

- › Implemented real-time windscreen effects (rain, snow, dirt, cleaning) for a runtime train simulator on Unreal Engine 5 (HLSL).
- › Worked in the R&D team led by Alain Kocelniack.

Inria | RESEARCH INTERN – GPU TERRAIN FILTERING

June 2022 – July 2022 | Grenoble, France

- › Designed a GPU height-map filtering algorithm for terrain generalisation (advisors: Joëlle Thollot, Romain Vergne).

Selected projects

Real-time soft-body simulation | C++ / CUDA

Dec. 2024

Spring-mass soft body accelerated on the GPU with CUDA.

Progressive raytracer | C++ / OpenGL

Dec. 2024

GPU raytracer with BVH traversal, shadows, reflections, refractions, environment maps and Cook–Torrance shading with a GGX microfacet distribution. Frame accumulation for progressive refinement.

Particle physics simulation | C++

May 2023

2D/3D simulation under gravity and a Lennard-Jones potential, with hand-written vector class and spatial data structures. Ran 8 000 particles over 400 000 steps in 20 minutes on a laptop.

Black Seas – multiplayer naval game | RUST / OpenGL, TEAM OF 5

Feb. – Apr. 2023

From-scratch engine; rendering, terrain, lighting, post-processing.

Contact

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🌐 briceprs.github.io ↗

Skills

Languages

C++, Rust, C, Python, GLSL, HLSL

Graphics & GPU

OpenGL, CUDA, WebGL, Unreal Engine 5, Shadertoy

Tools

Git, L^AT_EX, Linux

Familiar

Java, Shell, Assembly

Languages

French

Native

English

C1 – TOEFL 100 pts

Japanese

A1

Education

ENSIMAG (M.Eng.) | ÉCOLE NAT. SUP.

D'INFORMATIQUE ET DE MATH. APPL. – GRENOBLE INP

2021 – 2024 | Grenoble, France

Institute of Science Tokyo | ACADEMIC

EXCHANGE (FORMERLY TOKYO TECH)

Mar. – Aug. 2024 | Tokyo, Japan

Lycée Notre-Dame de Sion | CLASSES

PRÉPARATOIRES MPSI/MP

2018 – 2021 | Marseille, France

Interests

Climbing & bouldering — since 2011, with a few competitions. Also a keen hiker.

Computer graphics, physically based rendering, real-time simulation, Shadertoy.